

Measurement Based GaN-Si Thermal Simulation Model

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Abstract

A standardized thermal model for GaN-Si devices has been defined and agreed across the various MACOM sites. The model definition is intended for use with thermal simulation packages such as Ansys and Symmic. Layer structure and thermal parameters are defined through a combination of measurement analysis and standard material properties where appropriate.

I. INTRODUCTION

Gallium Nitride (GaN) is an attractive material for manufacturing high-frequency high-power devices for use in radar and mobile communications applications. Compared with GaAs and Si technology, GaN offers a wider bandgap with higher breakdown fields, higher power densities at high frequency, and lower on-resistance. Such high power densities, however, result in significant Joule heat generation in these devices making thermal management a crucial aspect in device package design. Measurements of the correct device temperature are also crucial for lifetime testing, where device performance is monitored as a function of time and temperature. [1]

In order to establish a standard and consistent method for peak temperature definition a project was undertaken with the University of Bristol to define a measurement verified Ansys thermal model for MACOM GaN-Si products. The team at the University of Bristol led by Martin Kubul are recognized as world leaders in the area of GaN device operation, reliability and modeling. The research group uses a combination of Raman thermography and Finite Element Method (FEM) thermal simulation using Ansys to establish an accurate measurement based thermal simulation model. The deliverables from MACOM were for a fully parameterized model which can be configured for various die geometries, attach methods, packaging and pcb configurations.

The Bristol approach to thermal analysis utilizes both measurement and simulation in order to extract peak internal die temperatures. The second phase of the project was to effectively replace the complex and expensive Raman measurement with tools available to MACOM such as Microsanj Thermoreflectance measurement and QFI infrared measurement. Raman is still regarded as the industry standard and most accurate method of thermal measurement and as such an occasional "calibration" measurement is still proposed particularly for new epi materials or significant process changes.

The Ansys simulation software uses an FEM approach to solving complex thermal problems. This approach has the advantage that detailed models can be computed, accurately representing the geometry of large devices [2]. The Bristol model is intended to be suitable for engineers to quickly generate a geometry and mesh for a particular die and package arrangement using a measurement verified model without requiring the overhead associated with setting up a complex Ansys simulation from scratch.

Many of the thermal parameters associated with the GaN-Si thermal stack are well defined, however careful definition of the process dependent transition layers are key to determining an accurate thermal simulation model. A physically accurate geometry is created in Ansys to replicate the Raman measurement setup. Based on the Raman measurements in the channel region and at the die corner it is possible to fit the transition layer and die attach parameters such that the model accurately reflects the Raman measurement. 3x 2.04mm and 3x 8.16mm devices were measured in order to provide a good model fit. Subsequent model verification was completed by measuring a 400um device which was not part of the original work. The Bristol defined Ansys model was found to accurately predict the device temperatures.

II. MEASUREMENT SYSTEMS

A. RAMAN

The Physics laboratory in Bristol have both Raman and IR thermal capabilities. Measurements on the Raman system are taken with 50x objective, and a 532nm laser on a spot size of 0.5um diameter. The method measures a Raman shift due to temperature in the channel area. Measurements are taken at 0.5um from the centre of the field plate. Raman shift represents average temperature over 3D volume of GaN as shown in Fig.1

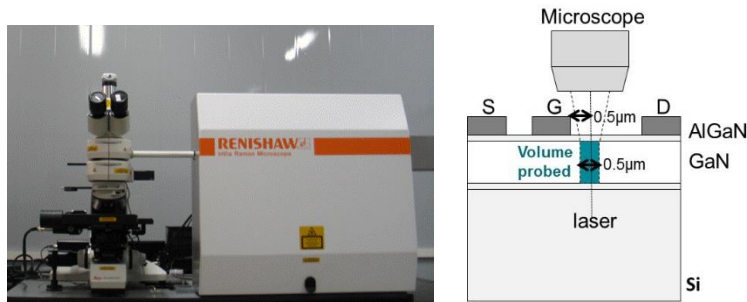


Fig.1. Raman thermal measurement system

B. MICROSANJ

The Microsanj NT200 series is the world's first commercially available thermoreflectance imaging analyser. This is a non-invasive visible light thermal imaging system with sub-micron spatial resolution, and time resolution of 100ns for transient analysis. Thermoreflectance imaging exploits the change in material reflectivity due to a change in temperature. This thermoreflectance technique uses a probing light source to detect the change in magnitude of reflected light as opposed to measuring the signal that is being emitted from the device due to black body radiation. Because of this, the probing light can be pulsed to measure the temperature at specified time delays with respect to the biasing pulse. This can also be done at cryogenic temperatures since we are not limited by photons emitted by blackbody radiation. The Microsanj system is inherently pulsed and is capable of producing a full transient characteristic response for a device.

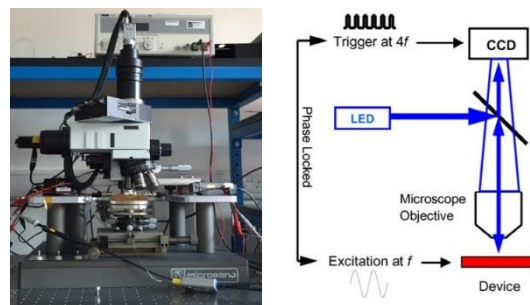


Fig.2. Microsanj thermal measurement system

A. QFI INFRARED

Infrared thermal microscopy involves the measurement of naturally emitted infrared radiation from the surface of a semiconductor device, such as an MMIC. Materials on the surface of the semiconductor device can vary in ability to emit infrared radiation. A ratio known as 'emissivity' is used to characterise the efficiency of a surface to emit infrared radiation, compared to a black body source. Accurate emissivity mapping is crucial to enable the actual surface temperature of the powered device to be obtained [3]. Quantum Focus Instruments Corporation (QFI) designs and manufactures advanced failure analysis microscope systems and temperature measurement microscope systems for the semiconductor industry.

II. MEASUREMENT CORRELATION

A detailed examination of the Ansys model is considered in section III, however in this section the initial measurements are correlated from Raman, Microsanj and Infrared, and compared to the Ansys model results. 3x 2.04mm and 3x 8.16mm die from WN031 were mounted on an AC360 package using AuSn preform. These devices were measured to help fit the Raman model and were subsequently measured on Microsanj as well as Infrared measurements at Bristol, Lowell, and the QFI facility. The infrared measurements use a 7um spot size and this is represented in the Ansys model as an average temperature over the surface at the gate region. Due to the transparency of GaN, the Microsanj measurement location is taken to be at the AlGaN-Si boundary. Fig.3 illustrates the model location for temperature extraction.

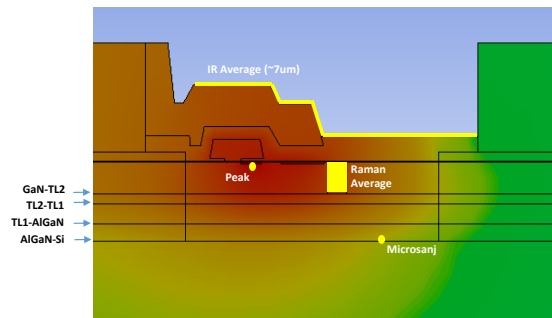


Fig.3 Thermal model image illustrating measurement locations

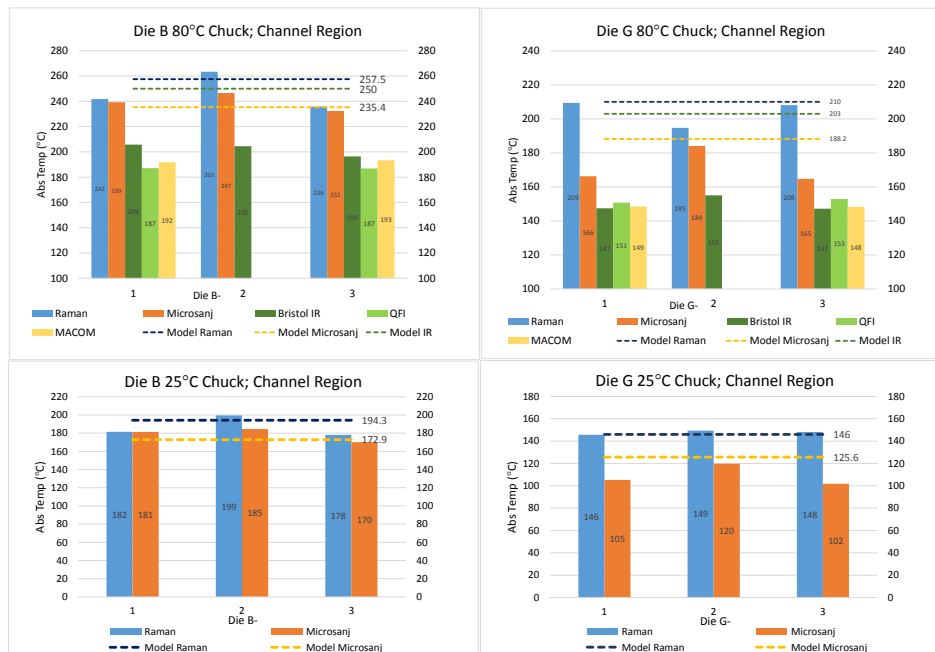


Fig.4. Measured and modelled temperatures for for 2.04mm (Die G) and 8.16mm (Die B) devices at 4W/mm

The graphs in Fig.4 show a comparison of the measurement results for the various systems, with the Ansys modeled temperature being represented by the dashed lines. The infrared measurements from the 3 sites yielded similar measurements, however it was not possible to directly correlate the infrared thermal measurements to the Ansys model. A similar result is reported in literature regarding infrared measurement and model correlation [2]. The Microsanj measurements and model predictions were found to be in reasonable agreement. Microsanj is fundamentally a pulsed based system which can generate a full thermal transient curve for a DUT. While a steady state situation can be replicated with a suitably long pulse, this introduces significant measurement challenges due to thermal expansion and movement. A method was developed to best facilitate this steady state measurement and recent system improvements in the form of a new piezoelectric autofocus unit with x,y,z correction have greatly improved the measurement experience and resulted in a more consistent correlation between Microsanj and Ansys model for subsequent measurements.

III. ANSYS THERMAL MODEL

A. MODEL DEFINITION

Although the Raman measurement is taken at a point reasonably close to the field plate edge, the true peak temperature can be significantly higher than this. It is therefore necessary to adopt a combined measurement/modelling approach, where the model is fitted to the measurement, facilitating peak temperature extraction directly from the model. This thermal model was established by the Bristol group based on Raman measurements and physical attributes of the device and epi provided by MACOM. The STEM images shown in Fig.5 were used to define the physical geometry for the model.

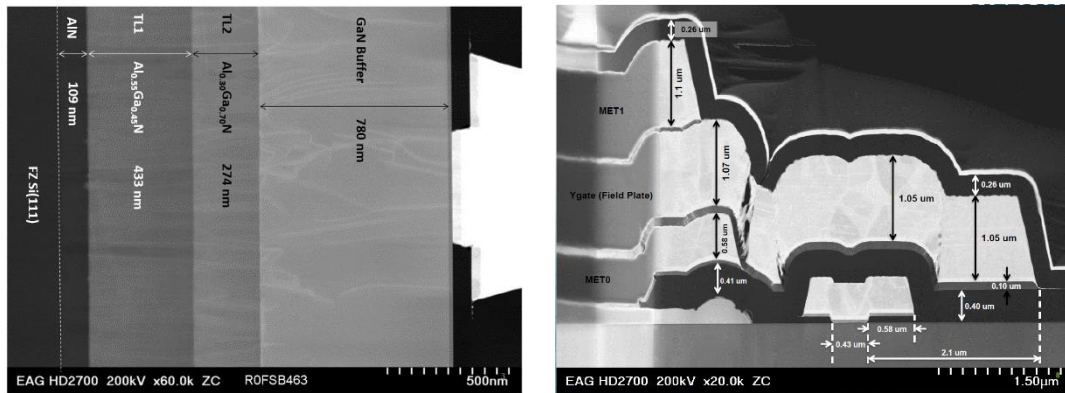


Fig.5 STEM image for WN024E007-05 - IQE Wafer 917-7

A Silvaco Atlas 2D drift diffusion model established from previous work was used to define the Joule heating distribution in the channel region [2]. The Ansys model heater elements have been defined under the gate and source field plates with values 62.5% and 37.5% respectively as shown in Fig.6. This figure also illustrates the agreed definition of peak temperature for use within the company. Using the peak temperature from the model to define a thermal resistance seemed unreasonable. It was therefore deemed more appropriate to define a 'gate foot average' temperature, to be taken at the centre of the gate stem to AlGaN boundary, and averaged over 50% of the gate width. This definition facilitates a more consistent correlation to other simulations packages such as symmic. It is also the location at which device failures are most likely to occur e.g. gate sinking.

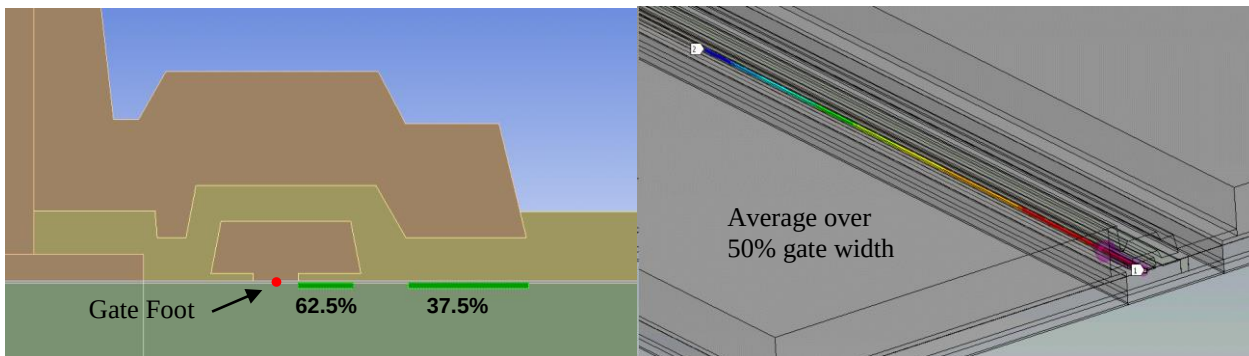


Fig.6 Joule heating distribution for Ansys model and 'gate foot average' temperature definition.

The full thermal stack definition for the model is illustrated in Fig.7. The thermal conductivity values reported in previous work were used, including temperature dependencies for both GaN and Silicon. The transition layers between GaN and Silicon (TL1, TL2 and AlN) are modelled using the same thermal conductivity value. This value along with the die attach thermal conductivity is adjusted in the model to fit the measured temperatures. Both these parameters need to be obtained from measurement since they can be affected by the fabrication process and epitaxial growth conditions. The die solder thermal conductivity was adjusted to fit the measured temperature at the corner of the die. Also included in the model is a

package to heatsink transition layer (Heatsink TL). This layer represents the thermal discontinuity at the base of the package, and needs to be considered for best model/measurement correlation. This layer can also be tuned for a particular fixture arrangement. For this epi definition, the parameters associated with the die are fixed to that shown in Fig.7. Package and die attach parameters should be modified to match the thermal stack under investigation.

* - Temperature Dependent Kth		
Nitride 10.4W/mK		
	1.18um	
	0.385um	0.57um
	0.411um	0.0777 um Nitride
0.016um	AlGaIn	10 W/mK
0.8um	GaN	112.72 W/mK* @150°C
0.27um	TL2	12 W/mK
0.425um	TL1	12 W/mK
0.1um	AlN	12 W/mK
50um	Si	94.52 W/mK* @150°C
0.5um	Au	315 W/mK
25um	Die Attach	30 W/mK
254um	Cu	380 W/mK
1016um	CuMo	210 W/mK
254um	Cu	380 W/mK
50um	Heatsink TL	2 W/mK

Fig.7. Thermal stack definition for GaN-Si using IQE epi on flange package

B. MODEL SCOPE

To best facilitate thermal simulation set-up in Ansys, a number of parameters were identified and built into the model by the Bristol team, such that geometry and mesh generation is automated within the defined model boundaries. In terms of die geometry, the following parameters are available: gate width, number of gates, die thickness, die attach thickness, die size. Simulation set-up allows parameters such as power density, base temperature and thermal conductivity values to be easily edited for a particular simulation. The model includes two separate packaging scenarios which can be connected as required. These consist of a 3 layer metal slug suitable for flange type packages or QFN's, and a laminate package with configurable vias. The Ansys view of the blocks is shown in Fig.8.

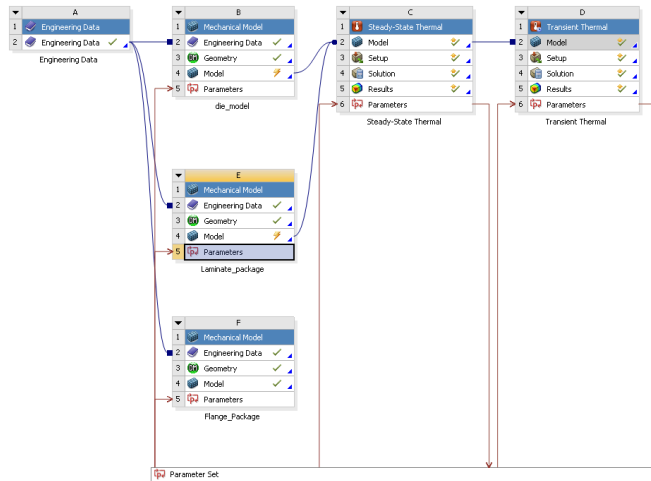


Fig.8. Ansys view of model configuration blocks

Meshing within the model uses a number of different strategies to achieve the desired accuracy while keeping the overall mesh to a computationally efficient size. For the base set-up of a single die, 2 planes of symmetry are exploited to reduce the overall problem size by a factor of 4. Fig.9 shows how the mesh density increases towards the die and channel regions. Peak temperatures are extracted from the centre of the die and therefore the central gate geometry and mesh is of higher density than the outer gates. The model also facilitates transient thermal simulations.

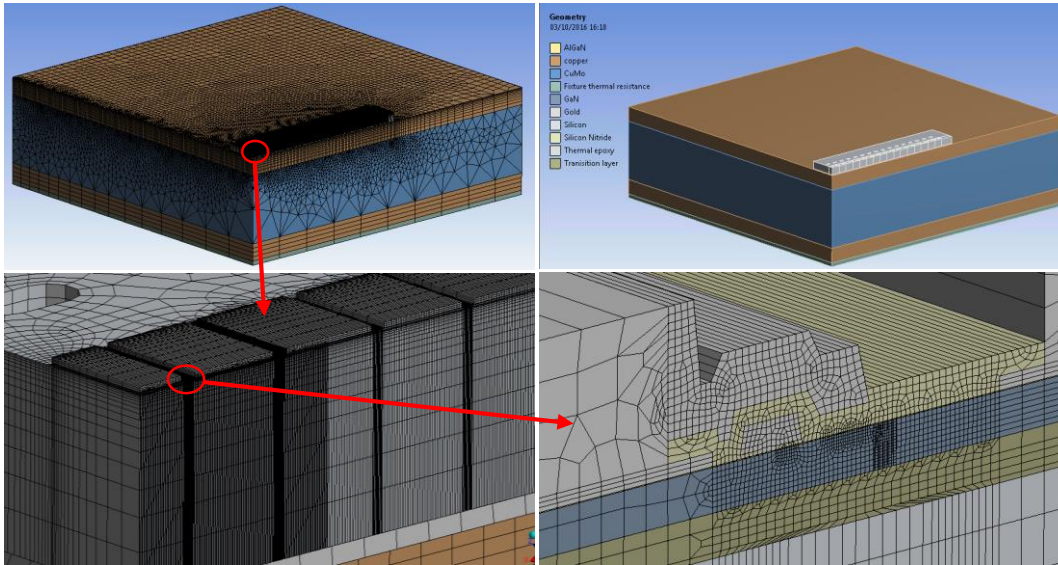


Fig.9. Ansys Meshing and material definition

III. VERIFICATION OF THE MODELING APPROACH

A 4x100um GaN-Si test structure mounted on a Kovar block and attached to an AC360 package was used to provide a further verification point for the modeling approach. This structure was built and tested on Raman and Microsanj, and simulated using the Ansys thermal model. The results summarized in Fig.10 show good correlation between both measurement techniques and the model. Further improvements to the Microsanj system are yielding better and more consistent results and provide good confidence that using a combination of Ansys model and Microsanj measurement will give a good indication of peak device temperatures. Raman measurements are still considered the ultimate reference so it is recommended to verify measurements on occasion using the Raman system, particularly if there is a change in epi or processing.

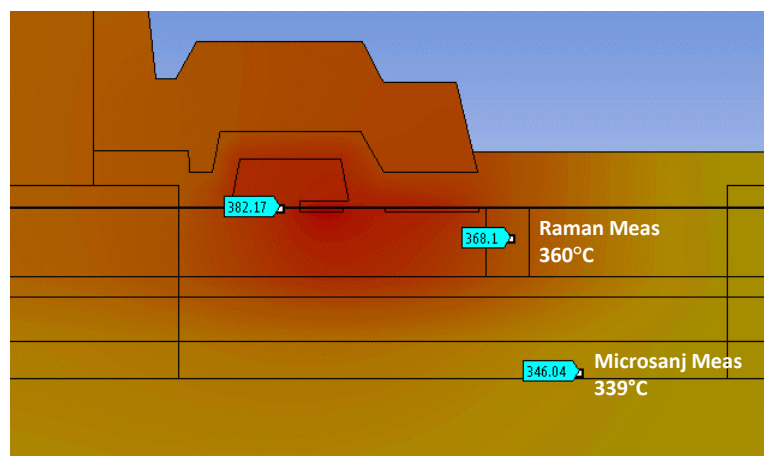


Fig.10. Model verification using 4x100um structure on Kovar and AC360 measured at 4W/mm and 150C baseplate temperature.

V. CONCLUSION

A measurement based GaN-Si thermal model has been established and verified. The die epi layer dimensions and gate structure have been measured from STEM images of equivalent devices. Material thermal conductivity values for the transition layers are extracted through a combination of Raman measurement and model fitting. Standard established values are used for other materials such as GaN and Si. A number of devices have been measured on Raman, Microsanj and QFI systems for comparison with the Ansys model. The model was fitted based on Raman measurement data, and shows good correlation with Microsanj measurements. It was not possible to correlate the infrared measurements to the Ansys model. A thermal die stack and heater element set-up is defined. A definition of peak temperature has also been agreed and can be extracted for a particular device through a combination of Microsanj measurement and Ansys simulation. The Ansys model has been set-up with a number of parameters to quickly simulate a particular die and package geometry. The modelling approach has been further verified on a different structure with good results.

ACKNOWLEDGEMENT

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